

Compare Results

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Total Changes

452

Content

36Replacements

22Insertions

175Deletions

Styling and Annotations

92Styling

127Annotations

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(Sennrich et al., 2015, 2021), which contains inflectional and derivational morphology for 15 languages. For our experiments, we only consider languages with Latin script and selected 10 languages of different language families. To annotate the separation of inflection suffixes and stem, we use MorphyNet’s inflectional information, where entries for an inflected form list the lemma, the morphological features and the canonical representation of the morphological segmentation (cf. table 1).

Some entries in the data set do not correspond to modern standard spelling (for example *poynnted* as English verb); thus we applied a filtering step based on two conditions: first, the lemma of the word needs to occur in a [dictionary](https://www.dict.cc)¹ (<https://www.dict.cc>) and second, the inflected word form needs to occur at least once in a text corpus for the respective language. For this purpose, we obtained a Wikipedia dump for every language. The filtering is designed to be rather conservative such that the word forms are valid forms of contemporary language, which is important when assessing the impact of the segmentation quality, where we want the test set to be as clean as possible. Table 8 (in the appendix) shows the number of entries after the filtering.

Additionally, the Wikipedia data is used to get an idea about a word’s frequency. While the frequencies in this text corpus do not correspond to those in the pre-training data, they still allow to approximately distinguish between high-frequency and low-frequency words.

1 Separation of Stem and Inflection

In this first experiment, we apply a linguistically-oriented criterion and study whether and how inflection suffixes are separated from the stem. We start from the hypothesis that a clean separation of inflectional suffixes allows for a better representation with regard to generalization due to separating the lexical content in the stem from the morphosyntactic information in the inflectional parts.

We define five categories, as illustrated in table 2, to describe the segmentation status of a word. Given the gold analysis, we compare how the word is segmented in the LM. The five categories are defined as follows:

- EXACT: the word is split into exactly two parts, the stem and the inflection suffix
- SINGLE: the inflection suffix consists of one piece; the stem is further split
- CONCAT: the inflection suffix consists of several pieces; the stem is or is not further split
- OVERLAP: there is no clear separation between the stem and the inflectional suffix

As we ignore the lemma part of the gold segmentation, there are no problems with irregular verb forms where the stem changes between lemma and inflected form. Many languages only have one suffix part, others like Finnish or Hungarian can have more. In the case of several suffixes, we only consider words where the concatenation of the suffixes in the gold segmentation also corresponds to the right side of the inflected word, but not forms like (ES) *abrámonos* → *abrir*—*amos*—*nos* (let’s open up) where the suffixes are represented in the canonical form and thus can deviate from the surface form.

The GPT segmentation was obtained for the target word without surrounding sentence context³.

Task: Verb Lemma Prediction

In this experiment, we investigate whether segmentation at the boundary between stem and inflectional suffixes has an effect on the task of predicting the lemma. As the frequency might be a relevant factor, we define 3 frequency ranges (cf. table 3) based on the observed frequency in the Wikipedia data. We compare verbs of the splitting category OVERLAP with verbs where inflection and stem are clearly separated (EXACT, SINGLE, CONCAT), with the hypothesis that verbs of the set OVERLAP should perform worse than verbs of the set NOT OVERLAP, as a clear separation between stem and inflection conceptually allows for a better generalization, in particular for words of lower frequency. We randomly select 500 verbs per group⁴; as common irregular verbs are typically listed in abundance in grammatical resources and thus are likely leaked in the pre-training data, we excluded the ten most common irregular verbs (according to gpt-4o) per language. Furthermore, we excluded verb forms that have the same

³We noticed that the segmentation of the word in sentence context can differ from the word in isolation, presumably due to the preceding space character which can influence segmentation, in particular for common words. To have similar conditions in the downstream task, we put the target word in parentheses such that there is no space on the left side.

⁴To obtain a diverse set, we use only one form per inflection paradigm if possible; otherwise, we use several forms per paradigm. Some settings still contain less than 500 entries.

surface form as the lemma, as the frequency of the word used as inflected form might differ considerably from the frequency of the form used as lemma.

We use the model gpt-4o with a relatively low temperature of 0.1 for a more stable outcome; the prompt is formulated in English for all languages:

Answer with one word.

2 Segmentation Consistency

The criterion in the previous section was based on linguistic well-formedness; here, we look at segmentation quality from the angle of consistency, which also aims at capturing generalization abilities, but is formulated more robustly. We pursue the question whether a consistent segmentation across the inflected forms of a lemma provides a better basis for the representation than an inconsistent segmentation. The underlying assumption is that an internally coherent representation of different surface realizations of the same word should result in an overall better representation of that word, and thus provide a better basis for generalization and the modeling of potentially unseen words. Table 4 shows some examples, ranging from a generally consistent representation of the stem part of the verb to a largely inconsistent segmentation⁵.

Ideally, a good segmentation should provide a consistent splitting of the stem part, with more necessary variation towards the end of the word. We use the Overlap Coefficient to measure the similarity between the sets of segments of two different verb forms, which is defined as the size of the intersection divided

by the size of the smaller one of the two sets:

$$\text{overlap}(A, B) = \frac{|A \cap B|}{\min(|A|, |B|)} \quad (1)$$

The scores range between 0 (no overlap) and 1 (perfect match). A particular characteristic of this metric is that if A is a subset of B, then the coefficient is 1: this has the effect of comparing rather the segments of the stem part while disregarding suffixes that add to the overall length of the word, assuming that the stem part does not change much, whereas we expect comparatively more variation in the suffixes. In contrast, the Jaccard index (ratio of intersection over union) might be less practical when the two compared forms are of different lengths, and we do not expect a subset to be similar.

Below are some examples for forms of the Italian verb sorprendere (to surprise), and the respective overlap scores between lemma and form:

lemma	form	overlap score	sorprendere
sorprenderebbe	1 sorprendere	sorprendiamo	
0.75 sorprendere	soprese	0.5	

The splitting in the first line is linguistically questionable, but the lemma's segments are an exact subset of the inflected form's segments, which is good in terms of consistency (overlap=1). For the other two words, the segments only partially match between the forms, and thus have a lower score. To obtain the overlap coefficient of an inflection paradigm, we computed the average of the overlap of every possible pair of forms. Table 2 shows an overview for all languages: for most languages, average overlap scores of 0.5 - 0.7 are dominant.

2.1 Paradigm Segmentation Overlap

In this experiment, we contrast verb forms from paradigms with high vs. low overlap coefficients: The underlying assumption is that the internal representation of verb forms with a less overlapping segmentation is sub-optimal as the forms cannot be well linked, whereas verbs with a high overlap coefficient are expected to be better connected within the paradigm.

⁵Note that in the example, the verbs follow the general rules of inflection, and that there are no major stem changes such as in e.g. go - went, which would lead to an even more inconsistent representation of the segmentation. Many verbs do have some sort of (semi-regular) surface variation in some dram atis ieren v inc ere ras en dram atis ierend v into ras end dram atis ieren vin ci rase dram atis ierten vince rast dram atis ierte vin cono ras est dram atis iert v inc ere b bero ras te dram atis iertet vin cess ero r asten to dramatize to win to speed

A further factor is the similarity of the segmentation of the inflected form to that of the lemma, i.e. the expected answer: a segmentation similar to the lemma is likely beneficial, thus further adding to the high/low overlap scenario. Note that we do not always have the full inflection paradigm of a verb at our disposition due to limitations of the dataset and our various filtering steps in the pre-processing; as inflection paradigm we thus define all observed forms of a verb lemma (with a minimum number of 5 forms per observed paradigm). We apply the following criteria to select 200 verbs per language.

- Average paradigm overlap: select verb paradigms with the highest/lowest average overlap coefficients per language
- Overlap to lemma: from those paradigms, select one form each with the highest/lowest overlap to the lemma (select at random if there are several forms with equal overlap)
- Frequency: additionally, we look at two frequency bands and consider forms with frequencies below 10 or above 500.

Based on this definition of high/low overlap, we select sets for the tasks of lemmatization and generation of inflected forms.

2.2 Positional Segmentation Differences

Here, we focus on consistent segmentation between the verb form and the lemma, and further assume that consistent segmentation at the beginning of the word, i.e. the lexical part of the word, is more important than at the end of the word, where the model is likely more robust due to observed variations with different inflections. We apply the following conditions to select verbs for three contrasting sets:

- Similarity: verb forms with a similarity to the lemma below 0.35 or above 0.7

⁶As want to study the extreme sides of high/low overlap, we opted for a smaller testset. Also, we did not consider English which has generally less forms per lemma.

Position of difference: verb forms with low similarity are grouped into subsets where the first subword token is the same for both words (`same1st`) or different (`diff1st`)⁷. *Frequency* : `formswitha frequencybelow(\low - freq")` or `above50(\high - freq")`

Table 7 shows the results: while we see the hypothesis that inconsistent segmentation at the beginning of a word has a negative effect confirmed, though not for all languages, we also have the somewhat surprising result that a matching first token, even with an otherwise low similarity, performs as well as the high-similarity group. One possible interpretation is that the relevant semantic information is already mostly contained in this first token.

3 Conclusion and Future Work

We proposed two criteria to capture the quality of subword segmentation in LLMs and evaluated to what extent words which score high or low for these criteria affect the performance of the LLM on a linguistic task for ten diverse languages. Both criteria are targeted at the generalization abilities of the language model; the first one is more linguistically inspired and aims at a clear separation of stem and inflectional suffixes, whereas the second one rewards consistent segmentation within an inflection paradigm. The design of the criteria is in principal language-independent, but requires language-specific information: morphologically annotated data for the first one, and information about sets of inflection paradigms for the second one.

The results of our experiments indicate that the subword segmentation does influence the behaviour of the model. In particular for the criterion of segmentation consistency, we could observe a better performance for the sets with higher segmentation overlap. In contrast, the morpheme-boundary criterion was found to be less suitable. With a view to lin-

⁷For all forms, we ensure that the first segment of the lemma can be a substring of the first segment of the form or vice-versa, in order to exclude forms that have a very different surface, for example `go-went`. This is a very weak condition;

guistic resources, the consistency-based criterion departs from a more minimal point, as no morphological analysis is needed other than knowing the inflection paradigm of a word.

The underlying segmentation is not only relevant for the representation within one language, but might also improve the multilingual competence of a model. Conceptually, when adhering to morpheme boundaries, the resulting segmentation can separate between lexical parts and functional components, which might benefit multi-lingual and structure learning; for instance, through supporting the learning of lexical equivalents of words sharing the same (or close) orthographic forms of the stem with different inflections, such as symbol -icEN, -ischDE, -iqueF R, -iczneP OL. Moreover, inflectional affixes contain context information such as tense or number, and an accessible and consistent representation can potentially contribute to the learning of syntactic structure across languages.

Finally, with view to the current efforts to include less-resourced languages into LMs, segmentation strategies that promote a consistent representation and maximize the generalization abilities are a relevant and interesting research field.

4 Limitations

In this section, we briefly discuss the limitations of the presented work.

Linguistic Tasks An obvious limitation are the simple linguistic downstream tasks that we used to evaluate the performance of the model. In our experiments, we mainly focus on predicting the lemma of a given verb form, which is arguably not the most exciting task, but has the advantage of being applicable to all languages in our data set. We extend this task to the generation of inflected forms based on lemma and morphological tags, which is more challenging and thus might be more affected by the underlying segmentation quality. A general issue with the generation task is that it is to a certain extent language-dependent due to the different morphological features per language, and consequently the op-

timal terminology to describe these features in the prompt. This makes this task likely more dependent on the prompt formulation than the lemmatization task.

In a certain way, both the prediction of lemmas and the generation of inflected forms are not necessarily natural tasks for the LLM. However, to better understand the impact of the underlying segmentation, we wanted a direct link between the investigated form and the linguistic task. This is more difficult to model in more complex tasks such as translation where more factors come into play.

Prompting We did not explore several prompt options, but used a simple and straightforward one. Similarly, we did not explore different prompt languages, but kept the English prompt for all investigated languages. In general, we are primarily interested in the performance difference between the sets of differently segmented words, but less in obtaining the best possible performance. Thus, to keep the conditions as simple as possible, we only looked at a zero shot scenarios in most experiments.

Non-Concatenative Morphology With regard to linguistic soundness in segmentation, a crucial factor that cannot be satisfactorily modeled by subword segmentation is non-concatenativity, such as irregular word forms (e.g. go – went), but also semi-regular variations such as an Umlaut in specific contexts, such as (DE) Apfelsg – Äpfelpl (applesg/pl). To fully capture these phenomena, one approach that has been proposed for both language modeling and machine translation is the representation of canonical forms in combination with morpho-syntactic information (e.g. Tamchyna et al. (2017), Antoun et al. (2020), Nzeyimana and Niyongabo Rubungu (2022), Jabbar (2024)), which however needs an additional step to generate inflected forms when generating, namely the generation of inflected forms based on the canonical representation in combination with the respective morphological features, which is not trivial.

In our study, we mostly ignored the problems of non-concatenative operations, in particular in the second part focusing on the segmentation consistency within a verb paradigm where phenomena such as stem changes between lemma and inflected form nec-

essarily lead to lower segmentation similarity. Our main reason is that regular segmentation strategies operating on surface words cannot handle morphological phenomena, and thus a linguistically sound modeling is out of reach with this method.

Languages and their Representation in the Training Data Finally, the amount of training data per language is also likely to have an influence on the segmentation quality for the respective languages, as suggested in Armengol-Estapé et al. (2022). With English making up the majority of the training data for GPT models, we would assume a distribution of subword tokens that best represents English, but not necessarily other languages, in particular if a language’s words differ considerably from English. While this is not a central point of our investigation of segmentation criteria in general, finding an optimal representation across languages is nonetheless a relevant factor that deserves attention in segmentation strategies for multilingual language models.

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A Data

Table 8 lists the number of inflected verb forms per language in our dataset.

Lang	Verbs	EN
23342 FR	57650	DE
21567 ES	15924	IT
49349 PT	27727	FI
22152 SV	14432	CS
20029 HU	37780	

Table 8: Overview of the number of verbs (inflected forms) per language after the filtering step.

Tags and Abbreviations

Table 9 lists the abbreviations used in MorphoNet’s tags and the respective feature names used in the prompt formulation, based on the documentation in magenta [https://unimorph.github.io/doc/unimorph-schema.pdf] (https://unimorph.github.io/doc/unimorph-schema.pdf)

V	verb	V.PTCP
participle IND	indicative	SBJV
subjunctive IMP	imperative	COND
conditional POT	potential	PST
past PRS	present	FUT
future SG	singular	PL
plural 1	first person	2
second person 3	third person	PFV
perfective IPVF	imperfective	PROG
progressive PRF	perfect	FORM
formal INFM	informal	

Table 9: Abbreviations and features used in the generation experiment.